

Sex hormonal effects on tendons and ligaments

Mette Hansen

Introduction

The biomechanical properties of tendons and ligaments differ between women and men. Furthermore, sex differences in injury risk are reported that depend on the type of sport, the training status and the age of the athletes. These observations have triggered speculation among athletes, trainers and coaches, as well as doctors and researchers, about a possible influence of sex hormones on the regulation of tendon and ligament structure, biomechanical properties and the ability to adapt and repair (Leblanc et al., 2017; Chidi-Ogbolu and Baar, 2018; Hansen, 2018).

The influence of sex hormones seems to vary between different tendons and ligaments, which may be attributed to their anatomical position, diverging tissue loading profiles and tissue structures. The composition of estrogen and androgen receptors may also vary, and the expression of estrogen receptors seems differentially influenced by the presence of estrogen. These observations underline the importance of taking into account an individual person's hormonal profile. Women's sex hormonal profiles differ markedly during their lifespan. In premenopausal, healthy and regularly menstruating women, female hormones fluctuate during each menstrual cycle and reach remarkably high levels during pregnancy. In postmenopausal women, however, the levels of estrogen and progesterone are negligible. In addition, the hormonal profile changes

with the use of hormonal contraception or hormone replacement therapy in young and postmenopausal women, respectively.

This chapter aims to give the reader a brief insight into current knowledge about the influence of sex hormones on ligament and tendon injury risk, structure, collagen turnover and biomechanical properties in young and aging women who are users or non-users of exogenous sex hormonal treatment (Figure 3.1).

Sex difference in tendon and ligament injury risk

The risk of sustaining anterior cruciate ligament (ACL) injury is 3- to 6-fold greater in female than in male athletes, even after taking into account socioeconomic, health and lifestyle background variables and level of sports participation (Renstrom et al., 2008). The annual incidence of ACL in female soccer and basketball players is around 5% and approximately 2- to 5-fold higher than seen in men (Prodromos et al., 2007; Renstrom et al., 2008). The risk is particularly high in high school-aged girls. Notably, an ACL injury can have serious consequences for the athlete. A retrospective study including 176 ACL patients previously engaged in organized football (soccer) showed that only half of them returned to activity and approximately one third gave up playing due to poor knee function or fear of additional injuries, which underlines the detrimental effect of this type of injury.

Chapter 3

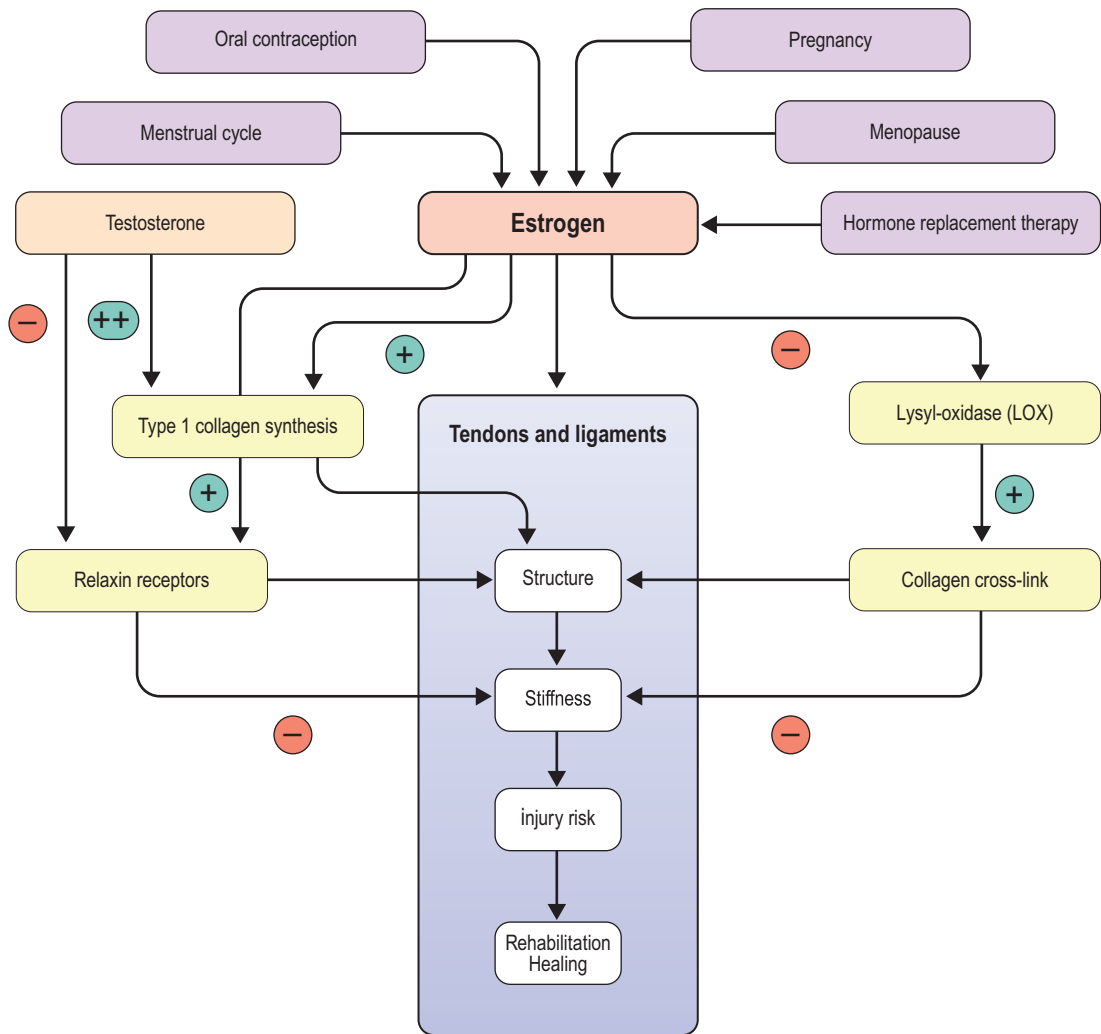


Figure 3.1

A schematic of possible modes of interaction between estrogen and T/L tissue suggested by the literature mentioned in this review. It also names common causes of changes in estrogen levels during the life of a woman. These mechanisms might affect different traits of tendon biology in diverse ways, thereby facilitating the changes observed in clinical studies.

Reprinted from Leblanc et al. (2017). Copyright © 2017 John Wiley and Sons. Used with permission.

Sex hormonal effects on tendons and ligaments

In contrast to ACL injury risk, the odds of sustaining Achilles rupture are lower in premenopausal women than in men (Huttunen et al., 2014). This sex discrepancy in risk disappears after menopause, where the incidence of Achilles rupture in women is similar to that of age-matched men as the level of estrogen is comparable between the sexes. Similarly, the risk of developing tendinopathy is lower in premenopausal women than in men (Cook et al., 2000), which indicates that female hormones play a protective role. In line with this, a higher prevalence of asymptomatic rotator cuff tears is observed after menopause, when the level of estrogen decreases markedly (Abate et al., 2014).

The sex difference in injury is probably explained by a complex interaction involving several risk factors (ligamentous laxity and size, limb alignment, neuromuscular control of motion and training status). The presence of sex hormone receptors in tendons and ligaments suggests that sex hormones may possibly affect tendon and ligament structure and biomechanical properties.

Sex differences in tendon and ligament structure and biomechanical properties

Ligaments such as the ACL improve joint stability. The stiffer a ligament, the lower the joint laxity and the risk of ligament rupture. In line herewith, joint laxity has repeatedly been reported to be larger in postpubertal women than in age-matched men (Deep, 2014), whereas no differences seem to exist between prepubertal boys and girls or between postmenopausal men and women. Furthermore, the sex disparity in ACL injury risk peaks in the teenage years, when sex hormonal differences are the largest (Renstrom et al., 2008). The latter suggests the presence of a sex hormonal effect on the tissue.

A stiff tendon may be beneficial from a purely performance-related perspective as it will transmit muscle force faster to the bone and store more energy than a compliant tendon. Furthermore, the maximal load before tendon failure is greater. Nevertheless, this benefit will come at the expense of a greater risk of muscle injury. Accordingly, tendon stiffness has been reported to be lower in young women than in men during voluntary isometric contractions using ultrasonography (Magnusson et al., 2007), and women seem to incur fewer muscle injuries than men (Edouard et al., 2016).

Sex differences in tendon and ligament biomechanical properties are partly explained by a generally larger tendon size in men than in women. Even so, the Achilles and patellar tendon cross-sectional area (CSA) is larger in young male runners than in equally trained young female runners when taking their training status into account and when adjusting for body weight differences (Magnusson et al., 2007) (Figure 3.2). Furthermore, tendon CSA is reported to be larger in trained men than in untrained men, which is in agreement with some randomized controlled studies and cross-sectional studies showing a hypertrophic effect of training on patellar and Achilles tendon CSA (Couppé et al., 2008; Kongsgaard et al., 2007). By contrast, we observed no difference in tendon CSA between untrained and trained female runners who had been running >40 km/week for the past 5 years (Magnusson et al., 2007). In agreement herewith, no change in Achilles tendon CSA was observed after 34 weeks of running (in total 43 hours) in previously untrained women (Hansen et al., 2003). Nevertheless, new data from our laboratory show an increase in patellar tendon CSA after only 10 weeks of resistance training (Dalgaard et al., 2019); and, in experienced female handball players, patellar tendon CSA is larger in the

Chapter 3

dominant jumping leg than in the contralateral leg (Hansen et al., 2013). The ability of the patellar and Achilles tendon to adapt to regular training may thus be reduced in women but not in men (Magnusson et al., 2007), or the loading threshold for adaptation may be enhanced in women. The lack of adaptation in female runners could also be attributed to hormonal disturbances related to low energy availability. Thus, amenorrhea caused by low energy availability has been reported among 60% of elite middle- and long-distance female runners and among 23% of elite female sprinters (Melin et al., 2019). These athletes are characterized by low estrogen, but also a low insulin-like growth factor I (IGF-I) level. The latter may impair the ability of their tendons to adapt to training because IGF-I has a stimulating effect on tendon collagen synthesis (Hansen et al., 2012).

The structural quality of tendons also appears to differ between young men and women. During

loading of single human patellar tendon collagen fascicles from young men and women, the ultimate stress before rupture is greater in fascicles from men than in those from women (Magnusson et al., 2007). The sex differences in fascicle mechanical properties may be related to structural differences.

The mechanical properties of both tendons and ligaments depend on collagenous fiber characteristics (type, density, diameter, orientation). The dry weight in tendons and ligaments is made up mostly of collagen (60–85%), of which type I collagen accounts for around 60% in tendons and up to 85% in ligaments. Nevertheless, the expression of type III collagen in the patellar tendon has been reported to be higher in premenopausal women than in men (Sullivan et al., 2009). Type III collagen is associated with greater elastic tissue flexibility, which is in agreement with the lower tendon and ligament stiffness observed in women. In addition, the patellar tendon dry mass and collagen content per tendon weight have been reported to be lower in women than in men (Lemoine et al., 2009).

The collagen content of ligaments and tendons is determined by the balance between the collagen protein synthesis rate and the collagen protein breakdown rate. However, tendinous collagen turnover is slow; moreover, recent data based on the carbon-14 ($[^{14}\text{C}]$) bomb pulse method suggests that in healthy humans the majority of the Achilles tendon collagen matrix is developed up to the age of approximately 13 years and not renewed after this age (Heinemeier et al., 2018). This observation runs contrary to previous findings, where the tendinous collagen synthesis rate was shown to increase in response to mechanical loading, as determined by the stable isotope technique measuring changes in isotopic enrichment in human tendon biopsies. A suggested explanation is that both exercise and hormones may influence a relatively small fraction of the

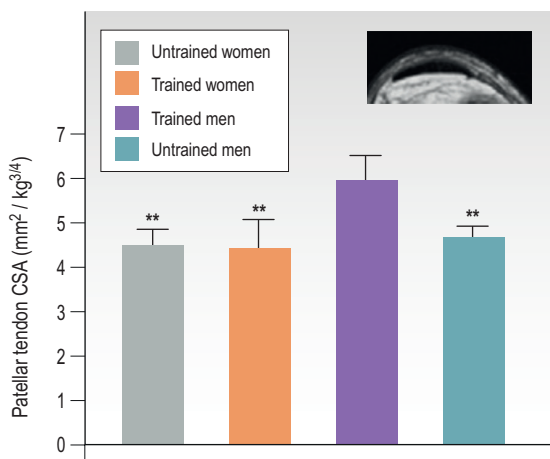


Figure 3.2

The magnetic resonance imaging-determined patellar tendon CSA for trained and untrained men and women normalized to body mass. Trained men had a greater CSA than untrained men ($P < 0.01$); however, note that trained women had a similar CSA compared with untrained women.

Reprinted from Magnusson et al. (2007). Copyright © 2007 John Wiley and Sons. Used with permission.

Sex hormonal effects on tendons and ligaments

collagen content that has a fast turnover. Still, this may be of importance for changes in the mechanical properties and in tendon and ligament size over time. In light of this, the observation of a lower tendinous collagen synthesis rate in young women than in age-matched men both at rest and in response to exercise at the same relative intensity is remarkable (Magnusson et al., 2007). A lower tendon and ligament collagen synthesis rate may reduce the ability to adapt in response to regular loading.

Improvements in tendon mechanical properties in response to training have been reported in the majority of human studies, even in studies where tendon CSA was not significantly changed. Yet the mechanical properties of tendons and ligaments are not only determined by the collagen content but also by the presence of intra- and intermolecular crosslinks, which enhance tissue stiffness. Collagen fibers can be crosslinked either non-enzymatically (Maillard reaction between a sugar and an amino acid) or enzymatically. Lysyl-oxidase (LOX) facilitates enzymatic crosslinking. In engineered ligaments, it has been shown that estrogen inhibits LOX and thereby reduces the relative stiffness (elastic modulus) (Lee et al., 2015). This observation suggests that, independently of any differences in collagen content within tendons and ligaments, sex differences in tendon stiffness and joint laxity may, at least in part, be explained by the higher estrogen level, which impairs tissue crosslinking.

Influence of estrogen on the structural and biomechanical properties of tendons

Female sex hormone fluctuations during the menstrual cycle

The sex hormonal and menstrual profile varies considerably between species, which questions

the transferability of results from animal studies to humans and the relevance of including animal data in a context focused on the effect of sex hormones on human beings.

In premenopausal women, 17- β estradiol is the predominant type of estrogen. Estradiol levels in women are several-fold higher than in men until menopause. In healthy premenopausal women, female hormones estradiol and progesterone fluctuate in a predictable pattern during the menstrual cycle (MC) (Figure 3.3). The mean cycle length is 28 days (range 20–45 days). During menses at the beginning of the MC (days 1–6), estradiol and progesterone are both at their lowest levels. Estradiol reaches a peak concentration in the late follicular phase (FP) (days 12–14) around the time of ovulation, when also the levels of luteinizing and follicle-stimulating hormones are peaking. In the luteal phase (LP), a second, lower rise (days 20–24) in estradiol is experienced. Progesterone gradually increases after ovulation and reaches the highest level during LP around day 19–24. Thereafter, a sharp decline in progesterone and estrogen occurs before a new cycle begins.

Use of oral contraceptives (OC) varies between countries, but in many Western European countries more than 50% of young women take OC. The primary reason for using OC is contraception, but other reasons include dysmenorrhea, controlling and manipulating the timing of menses, and reducing the symptoms experienced during a normal menstrual cycle. In OC users, the endogenous level of female hormones is suppressed, but daily OC consumption leads to increases in synthetic estradiol (ethinyl-estradiol) and synthetic progesterone (progestin) during the first hours after ingestion of the pill. Different types of OC are on the market. OC differ primarily with respect to the type of progestins, which bind to the progesterone

Chapter 3

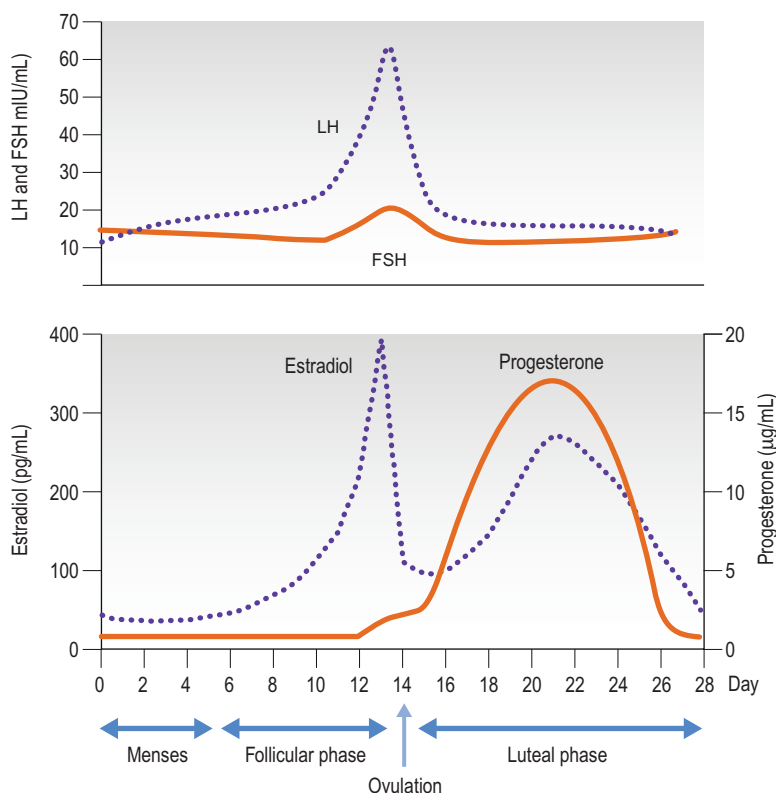


Figure 3.3

Hormonal changes during the menstrual cycle. Overview over the hormonal changes in estradiol and progesterone during the menstrual cycle in young women who do not use oral contraceptives.

Lower part reprinted from Laver et al. (2018). Copyright © 2018 Springer-Verlag GmbH, DE part of Springer Nature. Used with permission.

receptor with differential affinities and have differential androgenicity. Importantly, inter-subject variability in pharmacokinetic parameters after ingestion of similar types of OC is considerable. Therefore, women may react differently to OC administration depending on the type of OC as well as on unknown individual factors.

Estrogen receptors (α and β) are localized in the cell nucleus of tendons and ligaments and belong to the nuclear receptor superfamily. When the receptors are activated by ligand binding (estrogen) or other stimuli, they function as transcriptional regulators.

Activation of the ER α and ER β appears to induce both similar and opposing effects depending on the outcome parameter measured. The plasma membrane also contains

another type of estrogen receptor that can change intracellular cytosolic signaling pathways when activated by stimuli. Estrogen receptor distribution and the relative predominance of the different receptor isoforms vary within different tissue types, and the expression of receptors is influenced by the circulating level of estrogen.

Influence of estrogen on tendons and ligaments

It is difficult to study the isolated effect of estradiol in a human *in vivo* setting. Nevertheless, various efforts have been made to elucidate the effect of estradiol on tendons and ligaments and to clarify the effect of low versus high concentrations of estradiol during the menstrual cycle or between different groups of women.

Sex hormonal effects on tendons and ligaments

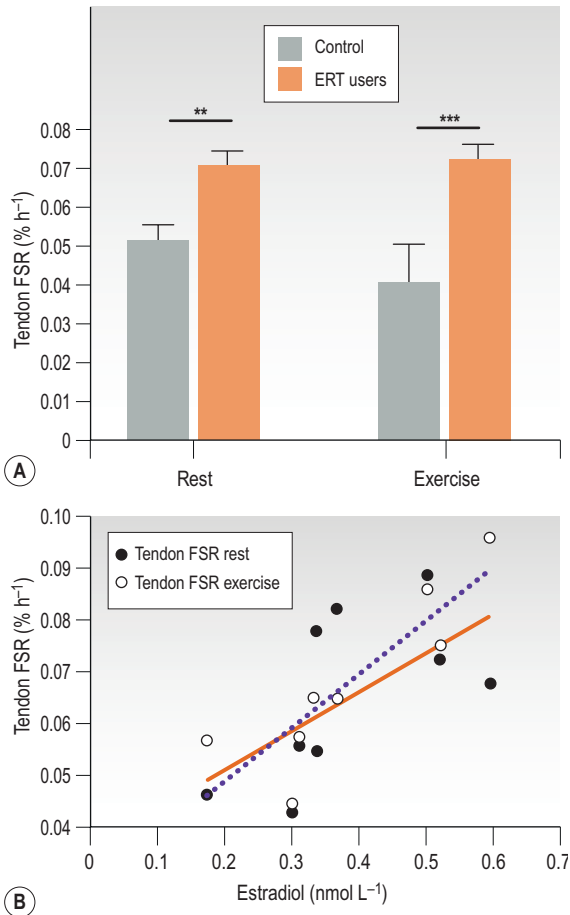


Figure 3.4

(A) Patellar tendon collagen fractional synthesis rates (FSR) at rest and 25 h after exercise in postmenopausal women who used estrogen replacement therapy (ERT) and postmenopausal women who did not use ERT (control); ** $P < 0.01$ and *** $P < 0.001$, unpaired t-test, control vs. ERT users. (B) The lines show relationship between tendon FSR and serum(s)-estradiol in ERT users at rest ($r^2 = 0.41$, $p = 0.06$) and postexercise ($r^2 = 0.80$, $P < 0.001$).

Copyright © 2009 The American Physiological Society. Used with permission.

during the menstrual cycle (Herzberg et al., 2018). Six of these studies provided quantitative data for a meta-analysis, which showed significantly greater knee joint laxity (0.36 mm) around ovulation compared with the early or mid-follicular phase, but this change was not significantly different from the knee joint laxity seen in the luteal phase. In the remaining six studies, the results of three studies suggested a greater knee laxity in the ovulation phase; one study concluded that the greatest knee laxity was observed in the luteal phase, and two studies found no significant difference between phases. The authors of the metaanalysis concluded that ACL laxity may be increased in the ovulation phase of the menstrual cycle, but more high quality studies are needed to confirm this finding. However, cyclic changes in genu recurvatum and general joint laxity have also been reported, which indicates a systemic sex hormonal effect (Shultz et al., 2010). The cyclic variation in knee laxity seems to be repeated in each menstrual cycle (Shultz et al., 2010). However, the magnitude and pattern of cyclic changes in knee laxity vary significantly among women, which may be due to inter-individual variations in hormonal changes during the menstrual cycle. In support of a direct association between estrogen and tendon stiffness, an inverse relationship between circulating estradiol concentration and tendon stiffness has been observed in female handball players (Hansen et al., 2013). Furthermore, the remarkably high level of estrogen during pregnancy is associated with consistently reduced ligament stiffness and enhanced knee joint laxity in the third trimester of pregnancy compared with 5–7 weeks postpartum (Charlton et al., 2001; Cherni et al., 2019).

Confirming the greater knee laxity in the late follicular phase (ovulation), systematic reviews suggest that the risk of sustaining ACL injury during

A systematic review from 2018 included 12 studies examining changes in knee laxity

Chapter 3

the menstrual cycle is lowest in the luteal phase and seems to peak in the late follicular phase (Herzberg et al., 2017) when estrogen, but not progesterone, is enhanced. The greater knee laxity associated with high levels of estrogen also seems to influence landing biomechanics and thereby the potential risk of non-contact ACL injury.

The mechanistic explanation for the rapid changes in ligament laxity and injury risk during the menstrual cycle is not fully elucidated. The effect of any explanatory factor needs to be rapid to explain cyclical variation in tissue laxity and injury. Evidence from animal studies shows that maximal load to failure is reduced in ACL in rabbits exposed to high estrogen levels. The collagen content probably does not change within days, but, as suggested previously, changes in the biomechanical properties of ligaments and tendons in adults are probably explained by changes in intra- and intermolecular crosslinking. In accordance with this, *in vitro* findings in engineered ligaments show that short-term (24 or 48 hours) exposure to a physiological concentration of estrogen decreases relative ligament stiffness and ultimate stress to failure, and this effect is linked to estrogen inhibiting the activity (61–77%) of the crosslinking enzyme LOX oxidase (Lee et al., 2015). If enzymatic activity is similarly reduced *in vivo* during exposure to high estrogen levels, this may explain the observations of greater knee laxity and the risk of sustaining ACL injury around the time of ovulation (Herzberg et al., 2017) as well as the reported lower tendon and ligament stiffness in women than in men.

Tendon and ligament structure and biomechanical properties may not be influenced by transient hormonal fluctuation during the menstrual cycle only. Over time, inter-individual variations in the sex hormonal profile and thus the accumulated exposure to estrogen may affect collagen turnover. A positive correlation

between circulating estrogen and human tendon collagen synthesis has been reported (Hansen et al., 2009a) (Figure 3.4). Furthermore, in elderly hysterectomized women who were long-term users of estrogen replacement therapy (ERT), tendon collagen synthesis was higher and patellar tendon relative stiffness lower than in age-matched postmenopausal women with very low estrogen levels (Hansen et al., 2009a). In the estrogen-deficient state, a lower tendon collagen synthesis and slower turnover rate might enhance the risk of introducing crosslinks, which will be further stimulated by LOX, which will no longer be inhibited by estrogen. In agreement herewith, estrogen deficiency in rats correlates with down-regulation of collagen turnover and alteration in collagen fiber orientation. In addition, in oophorectomized estrogen-deficient rats exposed to Achilles tendon injury, the tenocyte proliferation rate was lower than in sham-operated rats undergoing the same exposure. A reduced tenocyte proliferation and a lower tendon collagen synthesis rate in injured tendons or ligaments may potentially reduce the regenerative capacity and be non-beneficial. Furthermore, a protective effect of estrogen in premenopausal women may explain the lower risk of Achilles tendon rupture in premenopausal women than in men.

The different effect of high levels of circulating estrogen on the Achilles tendon and ACL rupture may be due to differences in tissue composition, loading profile and relative distribution and number of estrogen receptors (α and β). Furthermore, whereas a high estrogen level during the ovulation phase of the menstrual cycle in premenopausal women enhances joint laxity to a remarkable, potentially non-beneficial high level, lack of estrogen in the postmenopausal state might be non-beneficially related to a reduced type I collagen synthesis rate. In postmenopausal women, the content of type I collagen and the

Sex hormonal effects on tendons and ligaments

type I to type III+IV ratio are lower in the arcus tendinous fasciae pelvis than in women using hormone replacement therapy (HRT). The lower type I collagen content suggests a lower tensile tissue strength, which can be linked to increased susceptibility to anterior vaginal wall prolapse in elderly women. Furthermore, in active elderly women, lack of estrogen seems to reduce the ability to cope with Achilles tendon loading compared with active elderly women using HRT. In summary, both estrogen deficiency and a very high level of estrogen may have detrimental consequences for the mechanical properties of tendons and ligaments and injury risk.

Influence of oral contraceptives on tendons and ligaments

The effects of OC on tendon and ligament structure, mechanical properties and injury risk are not fully elucidated, but preliminary results indicate that OC use influences injury risk and the mechanical properties of ligaments and tendons. As previously mentioned, the risk of sustaining an ACL rupture is higher in active premenopausal women than in men. However, case-control studies indicate that OC use is associated with an up to 20% reduction in ACL injury risk compared with non-OC users (Rahr-Wagner et al., 2014). On the other hand, OC use is linked to an increased risk of Achilles tendinopathy (Holmes and Lin, 2006), whereas premenopausal women in general have a lower risk of experiencing Achilles injury than men until the menopausal transition. Explanations for these observations may be related to tendon and ligament stiffness, which have been reported to be higher in OC users than in female controls, but evidence for this remains sparse and inconsistent. A higher tendon stiffness in OC users fits well with observations of greater muscle damage and delayed recovery of muscle strength after eccentric, non-weight-bearing exercise compared with

the response in non-OC users (Minahan et al., 2015). Hypothetically, this observation may be explained by stiffer tendons in OC users, which enhances the tensile loading on the connected muscle-tendon junctions and contractile muscle filaments during muscle contractions. If it holds true that tendon stiffness is higher in OC users, this may be because the tendons experience no periodically enhanced estrogen level, as seen in non-OC users during the ovulation phase. In support hereof, knee joint laxity does not vary during the menstrual cycle as it would during the ovulation phase in non-OC users (Lee et al., 2014). Likewise, prospective data show no periodicity in non-contact ACL injury and ankle sprains in OC users (Agel et al., 2006). Still, it needs to be confirmed that the content of collagen crosslinks is higher in OC users because they experience no marked inhibition of the crosslinking enzyme LOX, like non-OC users in the ovulation phase. Alternatively, a higher ligament stiffness in OC users could be related to the fact that circulating and local concentrations of IGF-I are reduced and the tendon collagen synthesis rate lower in OC users than in controls (Hansen et al., 2009b). A human placebo-controlled study documented that local injections of IGF-I enhance the tendon collagen synthesis rate, which underlines the stimulating effect of IGF-I on tendon collagen synthesis (Hansen et al., 2012). Although the tendon collagen synthesis rate is reported to be lower in OC users, the collagen content does not seem to be lower. This indicates a lower collagen turnover, which will enhance the possibility of introducing and maintaining crosslinks within the collagen-rich tissues and thereby resulting in greater stiffness. However, the consequence of a lower collagen turnover may be an impaired ability to cope and adapt to a progressive increase in tissue loading during training. Also, the low level of IGF-I may impair rehabilitation after tendon and ligament injury since animal

Chapter 3

findings have documented that administration of IGF-I enhances tissue healing. Future studies are needed to clarify the potentially positive and negative effects of using OC on tendon injury risk, tissue mechanical properties, and adaptability to loading and rehabilitation after injuries.

Influence of androgen and relaxin on the structural and biomechanical properties of tendons

Androgen receptors are localized in human ACL tissue, which suggests that ACL tissue is androgen-responsive. The most abundant androgen in men and women is testosterone. However, the exact regulating role of testosterone on human tendons and ligaments remains elusive.

The level of testosterone is positively correlated with ACL stiffness, whereas estrogen and the estrogen-to-testosterone ratio are negatively correlated with ACL stiffness. An anabolic effect of testosterone on tendon collagen content and tissue stiffness is supported by human data showing a higher tendon collagen synthesis rate in young men than in women at rest (Magnusson et al., 2007). In addition, animal findings show that testosterone increases the collagen content in the prostate and hip joint capsule, and reduces the passive range of motion of the patellar tendon and ligament. Furthermore, *in vitro* findings have shown that high-dose testosterone administration to human cultured tenocytes increased the tenocyte number and changed the phenotype after short-term exposure. Therefore, the reported lower knee joint laxity and greater ligament and tendon stiffness in men than in women may be coupled to the approximately 7-fold higher testosterone level in young men than in young women. However, in normally menstruating premenopausal women, the testosterone level

fluctuates and is enhanced around ovulation, during which period increased knee laxity is observed. Still, in women, the testosterone level is low compared with that of men, and the findings suggest that the effect of estrogen on tendinous and ligamentous biomechanical properties in women is dominant. Nevertheless, testosterone may counteract estrogenic effects, as shown in mammary tissue. In support of the latter, animal data indicates that testosterone changes estrogen receptor α and β expression in opposite directions.

Testosterone may also have a positive influence on tendon and ligament adaptation to training. However, the human data basis is primarily based on cross-sectional data comparing men and women, showing that in response to acute exercise the tendon synthesis rate is higher in men than in women (Magnusson et al., 2007). Furthermore, after a 12-week alpine skiing program, the increase in patellar tendon stiffness was larger in elderly (aged 67 years) men (19%) than in elderly women (9%), even though training did not change tendon CSA significantly (Seynnes et al., 2011). One fact that suggests an androgenic rather than an estrogenic effect, is that the circulating estrogen level is comparable in elderly men and postmenopausal women, whereas testosterone levels are higher in men.

Similarly to estrogen, the potentially beneficial effect of testosterone on tendon and ligament injury risk probably depends on an optimal range of testosterone concentrations. In athletes who perform resistance training and use androgenic anabolic steroids, tendon stiffness and modulus are higher than in resistance-training athletes not using any androgenic anabolic steroids (Seynnes et al., 2013). However, tendon CSA was 15% higher in the athletic controls. The differences in patellar tendon morphology and mechanical properties suggest that abuse of androgens influences adaptation to training and

Sex hormonal effects on tendons and ligaments

collagen remodeling and entails higher stress on the tendon during loading. Supporting animal data shows that anabolic androgenic steroids reverse the beneficial effect of exercise on the Achilles tendon and result in inferior maximal stress values. In line with these observations, case reports indicate that abuse of testosterone or synthetic testosterone derivatives enhances the risk of tendon and ligament injuries and rupture (Kanayama et al., 2015). Superior gains in muscle mass and strength may further provoke an imbalance between training adaptations in the tendon and skeletal muscles, and explain the enhanced tendon rupture risk in abusers.

Relaxin is linked to collagenases activation, increased tendon laxity and enhanced risk of ACL injury, but also to improved ligament healing (Dehghan et al., 2014; Dragoo et al., 2011; Pearson et al., 2011). Interestingly, animal data suggests that testosterone downregulates the expression of relaxin receptors and thereby the effects of relaxin. By contrast, animal data shows that progesterone and high doses of estrogen enhance the expression of relaxin receptors. Therefore, testosterone, progesterone and estrogen may not only influence tendons and ligaments directly, but also tendon collagen turnover, collagen content and crosslinking by changing the responsiveness to relaxin.

Clinical summary

The risk of overuse and traumatic tendon and ligament injuries differs between women and men. Sex differences in tendon biomechanical properties, tendon morphology and tendon collagen turnover suggest that sex hormones play an explanatory role in this context. However, it is difficult to identify the isolated effect of the sex hormones on tendons and ligaments in human studies due to a complex interplay between internal and external risk factors, for example, aging, training status and nutrient intake, on top of hormonal

interactions. Nevertheless, both estrogen and testosterone in balanced physiological concentrations seem to be important for tendon health and physical function, whereas very low or high concentrations of endogenously or exogenously administered sex hormones may enhance the risk of injuries and lead to inadequate adaptation to mechanical loading. Remarkably, sex hormonal differences as well as sex differences in tendon and ligament injury risk are particularly distinct during sexual maturation in the teenage years, when the tendon and ligament collagen content is primarily established. Still, during adulthood and aging, marked changes in mechanical loading and in levels of endogenous and exogenous sex hormones seem to change the mechanical properties of tendons and ligaments, thereby changing the risk of acute and overuse injuries. Nevertheless, targeted preventive exercises are able to improve the biomechanical properties of tendons and ligaments, reduce joint laxity and reduce injury risk. Improved fitness and endurance capacity are important factors to reduce the risk of non-contact injuries, since fatigue may enhance risk by impairing coordination of movements. In addition, exercises aimed at improving hamstring activation and strength, as well as balance and proper landing and cutting techniques, may help to prevent ACL rupture (Bencke et al., 2018; Monajati et al., 2016; Renstrom et al., 2008).

From a future research perspective, it is important to point out that data addressing the theme of the present chapter remains sparse, and future studies are needed to establish in detail the effect of endogenous and exogenous sex hormones during the lifespan. To date, research has primarily focused on the ACL, the Achilles tendon and the patellar tendon. Future research may elucidate whether sex hormones influence tendons and ligaments differently depending on their localization and function.

Chapter 3

References

- Abate, M., Schiavone, C., Di Carlo, L. & Salini, V. (2014) Prevalence of and risk factors for asymptomatic rotator cuff tears in postmenopausal women. *Menopause*. 21: 275–280.
- Agel, J., Bershadsky, B. & Arendt, E.A. (2006) Hormonal therapy: ACL and ankle injury. *Med Sci Sports Exerc*. 38: 7–12.
- Bencke, J., Aagaard, P. & Zebis, M.K. (2018) Muscle activation during ACL injury risk movements in young female athletes: a narrative review. *Front Physiol*. 9: 445.
- Charlton, W.P., Coslett-Charlton, L.M. & Ciccotti, M.G. (2001) Correlation of estradiol in pregnancy and anterior cruciate ligament laxity. *Clin Orthop Relat Res*. 165–170.
- Cherni, Y., Desseauve, D., Decatoire, A., Veit-Rubinc, N., Begon, M., Pierre, F. & Fradet, L. (2019) Evaluation of ligament laxity during pregnancy. *J Gynecol Obstet Hum Reprod*.
- Chidi-Ogbolu, N. & Baar, K. (2018) Effect of estrogen on musculoskeletal performance and injury risk. *Front Physiol*. 9: 1834.
- Cook, J.L., Khan, K.M., Kiss, Z.S. & Griffiths, L. (2000) Patellar tendinopathy in junior basketball players: a controlled clinical and ultrasonographic study of 268 patellar tendons in players aged 14–18 years. *Scand J Med Sci Sports*. 10: 216–220.
- Couppé, C., Kongsgaard, M., Aagaard, P., Hansen, P., Bojsen-Møller, J., Kjaer, M. & Magnusson, S.P. (2008) Habitual loading results in tendon hypertrophy and increased stiffness of the human patellar tendon. *J Appl Physiol*. 105: 805–810.
- Dalgaard, L.B., Dalgas, U., Andersen, J.L., Rossen, N.B., Møller, A.B., Stødkilde-Jørgensen, H., Jørgensen, J.O., Kovanen, V., Couppé, C., Langberg, H., Kjaer, M., Hansen, M. (2019) Influence of Oral Contraceptive Use on Adaptations to Resistance Training. *Front Physiol*. Jul 2;10: 824.
- Deep, K. (2014) Collateral ligament laxity in knees: what is normal? *Clin Orthop Relat Res*. 472: 3426–3431.
- Dehghan, F., Haerian, B.S., Muniandy, S., Yusof, A., Dragoo, J.L. & Salleh, N. (2014) The effect of relaxin on the musculoskeletal system. *Scand J Med Sci Sports*. 24: e220–e229.
- Dragoo, J.L., Castillo, T.N., Braun, H.J., Ridley, B.A., Kennedy, A.C. & Golish, S.R. (2011) Prospective correlation between serum relaxin concentration and anterior cruciate ligament tears among elite collegiate female athletes. *Am J Sports Med*. 39: 2175–2180.
- Edouard, P., Branco, P. & Alonso, J.M. (2016) Muscle injury is the principal injury type and hamstring muscle injury is the first injury diagnosis during top-level international athletics championships between 2007 and 2015. *Br J Sports Med*. 50: 619–630.
- Hansen, M. (2018) Female hormones: do they influence muscle and tendon protein metabolism? *Proc Nutr Soc*. 77: 32–41.
- Hansen, P., Aagaard, P., Kjaer, M., Larsen, B. & Magnusson, S.P. (2003) Effect of habitual running on human Achilles tendon load-deformation properties and cross-sectional area. *J Appl Physiol*. 95: 2375–2380.
- Hansen, M., Boesen, A., Holm, L., Flyvbjerg, A., Langberg, H. & Kjaer, M. (2013) Local administration of insulin-like growth factor-I (IGF-I) stimulates tendon collagen synthesis in humans. *Scand J Med Sci Sports*. 23(5): 614–9.
- Hansen, M., Couppé, C., Hansen, C.S., Skovgaard, D., Kovanen, V., Larsen, J.O., Aagaard, P., Magnusson, S.P. & Kjaer, M. (2013) Impact of oral contraceptive use and menstrual phases on patellar tendon morphology, biochemical composition, and biomechanical properties in female athletes. *J Appl Physiol*. 114: 998–1008.
- Hansen, M., Kongsgaard, M., Holm, L., Skovgaard, D., Magnusson, S.P., Qvortrup, K., Larsen, J.O., Dahl, M., Serup, A., Frystyk, J., Flyvbjerg, A., Langberg, H. & Kjaer, M. (2009a) Effect of estrogen on tendon collagen synthesis, tendon structural characteristics, and biomechanical properties in postmenopausal women. *J Appl Physiol*. 106: 1385–1393.
- Hansen, M., Miller, B.F., Holm, L., Doessing, S., Petersen, S.G., Skovgaard, D., Frystyk, J., Flyvbjerg, A., Koskinen, S., Pingel, J., Kjaer, M. & Langberg, H. (2009b) Effect of administration of oral contraceptives in vivo on collagen synthesis in tendon and muscle connective tissue in young women. *J Appl Physiol*. 106: 1435–1443.
- Heinemeier, K.M., Schjerling, P., Ohlenschlaeger, T.F., Eismark, C., Olsen, J. & Kjaer, M. (2018) Carbon-14 bomb pulse dating shows that tendinopathy is preceded by years of abnormally high collagen turnover. *FASEB J*. 32: 4763–4775.
- Herzberg, S.D., Motu'apuaka, M.L., Lambert, W., Fu, R., Brady, J. & Guise,

Sex hormonal effects on tendons and ligaments

- J.M. (2017) The effect of menstrual cycle and contraceptives on ACL injuries and laxity: a systematic review and meta-analysis. *Orthop J Sports Med.* 5: 2325967117718781.
- Holmes, G.B. & Lin, J (2006) Etiologic factors associated with symptomatic achilles tendinopathy. *Foot Ankle Int.* 27: 952–959.
- Huttunen, T.T., Kannus, P., Rolf, C., Fellander-Tsai, L. & Mattila, V.M. (2014) Acute achilles tendon ruptures: incidence of injury and surgery in Sweden between 2001 and 2012. *Am J Sports Med.* 42: 2419–2423.
- Kanayama, G., Deluca, J., Meehan, W.P., Hudson, J.I., Isaacs, S., Baggish, A., Weiner, R., Micheli, L. & Pope, H.G. (2015) Ruptured tendons in anabolic-androgenic steroid users: a cross-sectional cohort study. *Am J Sports Med.* 43: 2638–2644.
- Kongsgaard, M., Reitelsheder, S., Pedersen, T.G., Holm, L., Aagaard, P., Kjaer, M. & Magnusson, S.P. (2007) Region specific patellar tendon hypertrophy in humans following resistance training. *Acta Physiol.* 191: 111–121.
- Leblanc, D.R., Schneider, M., Angele, P., Vollmer, G. & Docheva, D. (2017) The effect of estrogen on tendon and ligament metabolism and function. *J Steroid Biochem Mol Biol.* 172: 106–116.
- Lee, C.A., Lee-Barthel, A., Marquino, L., Sandoval, N., Marcotte, G.R. & Baar, K. (2015) Estrogen inhibits lysyl oxidase and decreases mechanical function in engineered ligaments. *J Appl Physiol.* 118: 1250–1257.
- Lee, H., Petrofsky, J.S., Daher, N., Berk, L. & Laymon, M. (2014) Differences in anterior cruciate ligament elasticity and force for knee flexion in women: oral contraceptive users versus non-oral contraceptive users. *Eur J Appl Physiol.* 114: 285–294.
- Lemoine, J.K., Lee, J.D. & Trappe, T.A. (2009) Impact of sex and chronic resistance training on human patellar tendon dry mass, collagen content, and collagen cross-linking. *Am J Physiol Regul Integr Comp Physiol.* 296: R119–R124.
- Magnusson, S.P., Hansen, M., Langberg, H., Miller, B., Haraldsson, B., Westh, E.K., Koskinen, S., Aagaard, P. & Kjaer, M. (2007) The adaptability of tendon to loading differs in men and women. *Int J Exp Pathol.* 88: 237–240.
- Melin, A.K., Heikura, I.A., Tenforde, A. & Mountjoy, M. (2019) Energy availability in athletics: health, performance and physique. *Int J Sport Nutr Exerc Metab.* 1–35.
- Minahan, C., Joyce, S., Bulmer, A.C., Cronin, N. & Sabapathy, S. (2015) The influence of estradiol on muscle damage and leg strength after intense eccentric exercise. *Eur J Appl Physiol.* 115: 1493–1500.
- Monajati, A., Larumbe-Zabala, E., Goss-Sampson, M. & Naclerio, F. (2016) The effectiveness of injury prevention programs to modify risk factors for non-contact anterior cruciate ligament and hamstring injuries in uninjured team sports athletes: a systematic review. *PLoS One.* 11: e0155272.
- Pearson, S.J., Burgess, K.E. & Onambele, G.L. (2011) Serum relaxin levels affect the in vivo properties of some but not all tendons in normally menstruating young women. *Exp Physiol.* 96: 681–688.
- Prodromos, C.C., Han, Y., Rogowski, J., Joyce, B. & Shi, K. (2007) A meta-analysis of the incidence of anterior cruciate ligament tears as a function of gender, sport, and a knee injury-reduction regimen. *Arthroscopy.* 23: 1320–1325.
- Rahr-Wagner, L., Thillemann, T.M., Mehnert, F., Pedersen, A.B. & Lind, M. (2014) Is the use of oral contraceptives associated with operatively treated anterior cruciate ligament injury? A case-control study from the Danish Knee Ligament Reconstruction Registry. *Am J Sports Med.* 42: 2897–2905.
- Renstrom, P., Ljungqvist, A., Arendt, E., Beynon, B., Fukubayashi, T., Garrett, W., Georgoulis, T., Hewett, T.E., Johnson, R., Krosshaug, T., Mandelbaum, B., Micheli, L., Myklebust, G., Roos, E., Roos, H., Sschamasch, P., Shultz, S., Werner, S., Wojtys, E. & Engebretsen, L. (2008) Non-contact ACL injuries in female athletes: an International Olympic Committee current concepts statement. *Br J Sports Med.* 42: 394–412.
- Seynnes, O.R., Kamandulis, S., Kairaitis, R., Helland, C., Campbell, E.L., Brazaitis, M., Skurvydas, A. & Narici, M.V. (2013) Effect of androgenic-anabolic steroids and heavy strength training on patellar tendon morphological and mechanical properties. *J Appl Physiol.* 115: 84–89.
- Seynnes, O.R., Koesters, A., Gimpl, M., Reiffberger, A., Niederseer, D., Niebauer, J., Pirich, C., Muller, E. & Narici, M.V. (2011) Effect of alpine skiing training on tendon mechanical properties in older men and women. *Scand J Med Sci Sports.* 21: S39–S46.
- Shultz, S.J., Levine, B.J., Nguyen, A.D., Kim, H., Montgomery, M.M. & Perrin,

Chapter 3

D.H. (2010) A comparison of cyclic variations in anterior knee laxity, genu recurvatum, and general joint laxity across the menstrual cycle. *J Orthop Res.* 28: 1411–1417.

Sullivan, B.E., Carroll, C.C., Jemiolo, B., Trappe, S.W., Magnusson, S.P., Dossing, S., Kjaer, M. & Trappe, T.A. (2009) Effect of acute resistance exercise and sex on human patellar tendon structural and

regulatory mRNA expression. *J Appl Physiol.* 106: 468–475.